



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to S.M.P FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

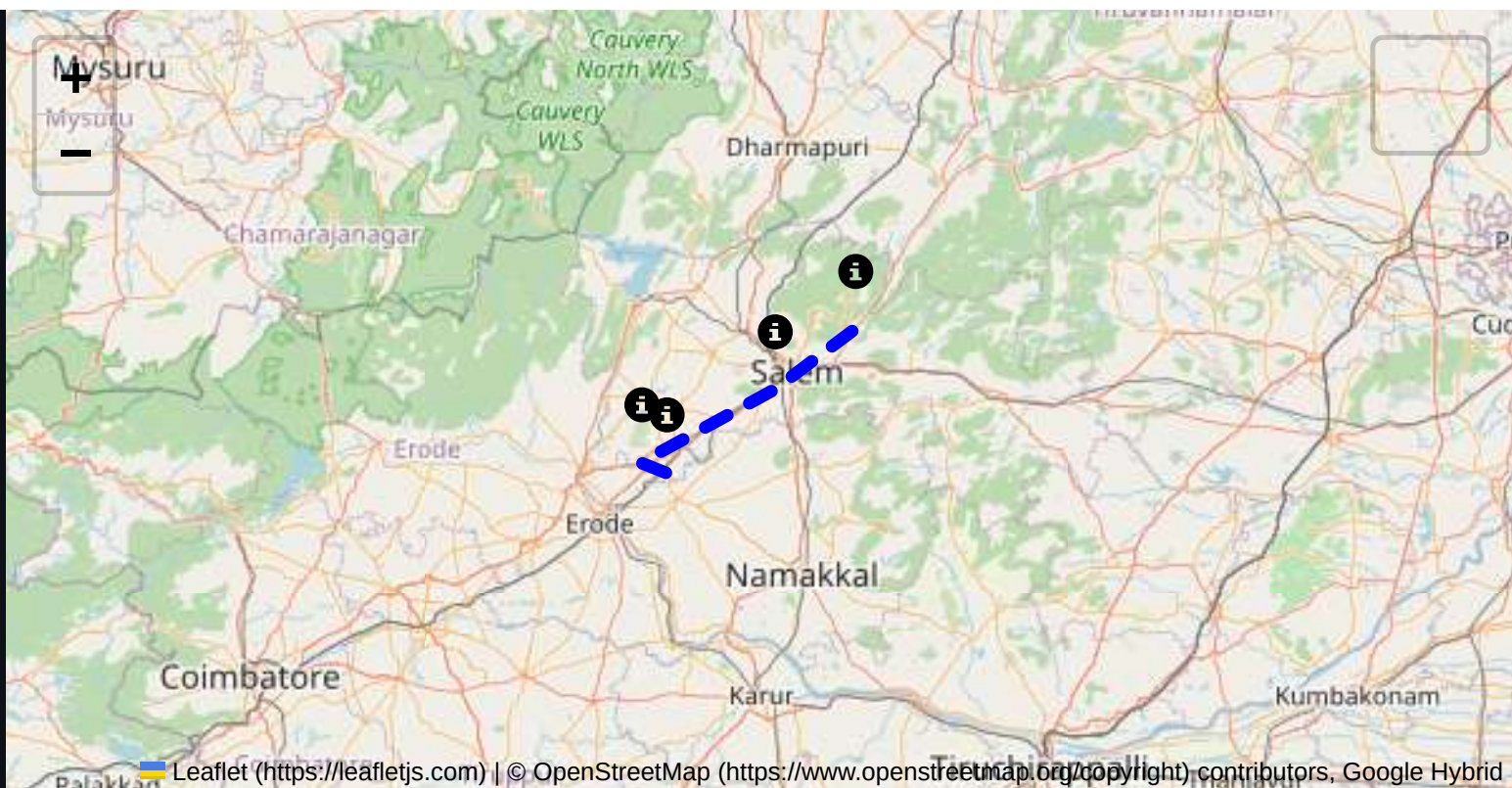
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

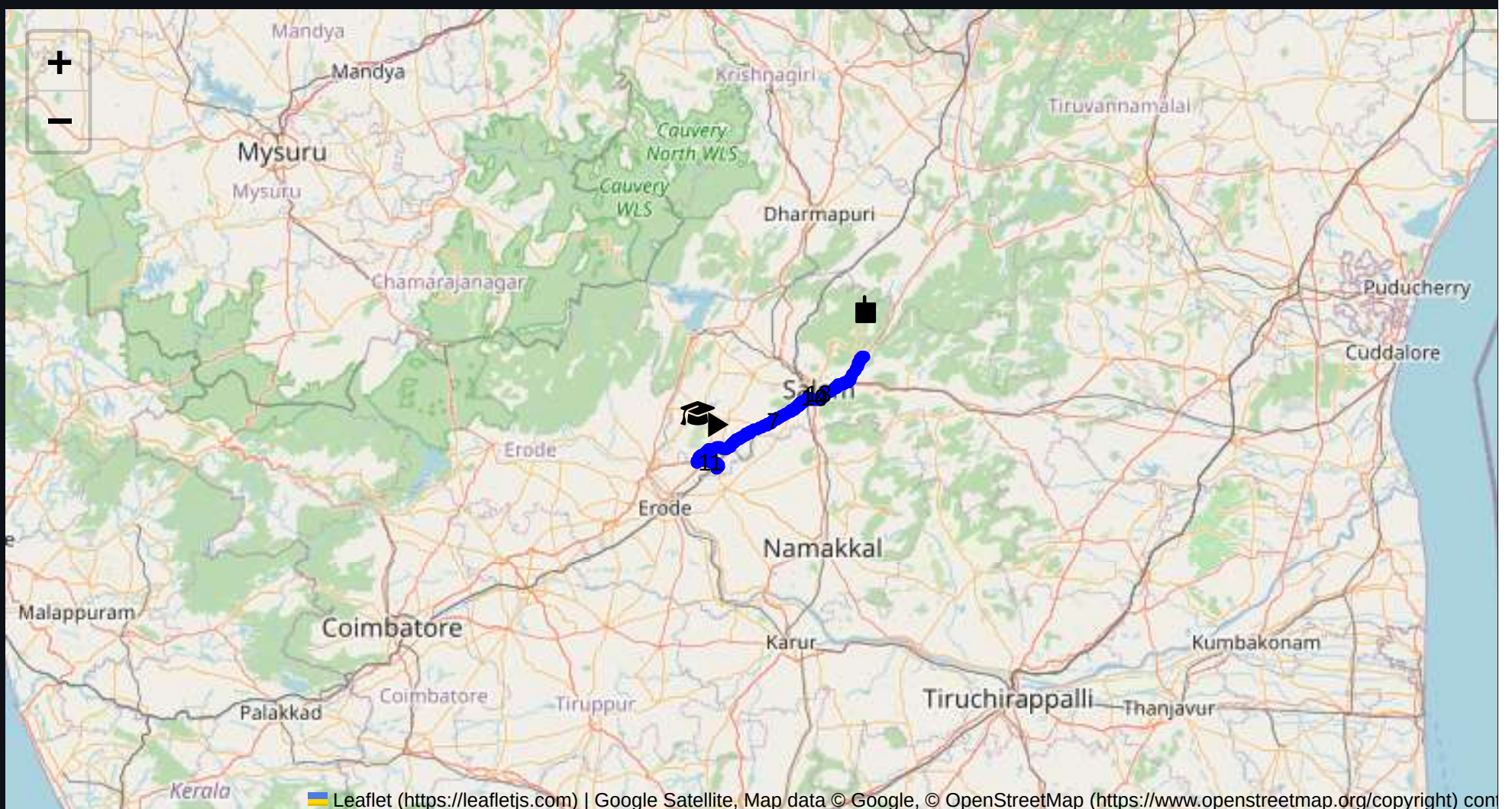
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 75.19 km
Estimated Duration: 1.6 hours
Adjusted Duration (Heavy Vehicle): 2.0 hours
Start: (11.4381, 77.8734)
End: (11.746118, 78.281718)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map: The route from Sangagiri to Kuppanur via Padaiveedu and Uthamasolapuram covers approximately 75.19 kilometers. It follows a path through several small towns and rural areas, eventually reaching the outskirts of the city of Salem, where you continue onwards to Kuppanur.

2. Typical Weather Conditions and Potential Weather-Related Hazards: Tamil Nadu experiences a tropical climate. The potential weather-related hazards include:

- **Monsoon Rains (June to September):** This can lead to slippery roads and localized flooding, especially in low-lying areas.
- **High Temperatures (March to June):** This can affect vehicle tires and increase driver fatigue.
- **Occasional Cyclones (October to December):** This can cause severe disruptions, including road blockages.

3. Traffic Patterns, Highlighting Peak Hours and Congestion-Prone Areas:

- **Peak Hours:** Typically between 8:00-10:00 AM and 5:00-7:00 PM.
- **Congestion-Prone Areas:** Salem city outskirts can experience high traffic congestion. Bhavani Main Road is also noted for heavy traffic, especially near towns.

4. Road Quality and Infrastructure Assessment:

- **Overall Quality:** The roads are a mix of well-maintained highways and smaller rural roads that may have potholes and uneven surfaces.
- **Infrastructure Issues:** Lack of adequate lighting and narrow road sections in rural areas can be challenging, especially at night.

5. Suggestions for Alternative Routes for Emergencies: In case of road blockages:

- **Via State Highway 86:** Offers a longer but viable alternative path.
- **Local Roads:** Knowledge of minor roads can help in bypassing specific congestion spots in emergencies.

6. Summary of Local Regulations Affecting Hazardous Material Transport: Tamil Nadu mandates permits for transporting hazardous materials, with specific routes often favored to minimize risks in densely populated areas. Night travel restrictions on certain routes may apply.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials:

- **Accidents:** Previous incidents have involved overturned trucks due to sharp turns or speeding, particularly in hilly areas near Salem.
- **Spills:** Fewer incidents but there's a history of chemical spills near industrial zones.

8. Environmental Considerations and Sensitive Areas:

- **Ecologically Sensitive Areas:** Include sections near reserved forests or wildlife sanctuaries where reduced speed limits and specific driving instructions may apply.
- **Pollution Concerns:** Ensuring vehicles meet emission standards is crucial to mitigate air quality impacts.

9. Analysis of Communication Coverage, Noting Potential Dead Zones:

- **Good Coverage:** Near towns and Salem.
- **Potential Dead Zones:** Remote rural areas might have sporadic mobile signal coverage.

10. Estimated Emergency Response Times for Different Route Segments:

- **Urban Areas (Salem vicinity):** 15-30 minutes.
- **Rural Sections:** Can range from 30-60 minutes, longer in remote areas due to poor road access.

11. Overall Summary of Risk Assessment: The route involves moderate risk primarily due to varied road quality, traffic conditions in urban areas, and potential weather impacts. Suitable precautions include checking weather reports, avoiding peak traffic times, and ensuring compliance with regulations on hazardous materials. Emergency planning should include alternative routes and ensure reliable communication means for drivers. Consideration of environmental and local regulatory guidelines is essential for minimizing risks during transit.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63266, 78.12699	15 KM/Hr
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.44900, 77.87406	30 KM/Hr
5	Turn	Medium	11.45358, 77.85781	30 KM/Hr
6	Turn	High	11.45353, 77.85697	15 KM/Hr
7	Turn	Medium	11.45853, 77.85139	30 KM/Hr
8	Turn	Medium	11.46307, 77.84941	30 KM/Hr
9	Turn	Medium	11.46316, 77.84921	30 KM/Hr
10	Turn	High	11.45542, 77.81725	15 KM/Hr
11	Blind Spot	Blind Spot	11.45631, 77.81668	10 KM/Hr
12	Turn	Medium	11.60942, 78.10056	30 KM/Hr
13	Turn	Medium	11.60965, 78.10070	30 KM/Hr
14	Blind Spot	Blind Spot	11.62145, 78.10740	10 KM/Hr
15	Turn	High	11.62130, 78.10776	15 KM/Hr
16	Blind Spot	Blind Spot	11.61217, 78.10439	10 KM/Hr
17	Turn	High	11.63297, 78.12749	15 KM/Hr
18	Turn	Medium	11.62826, 78.13337	30 KM/Hr
19	Turn	Medium	11.65632, 78.20220	30 KM/Hr
20	Turn	High	11.66380, 78.20648	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
21	Turn	High	11.67111, 78.23134	15 KM/Hr
22	Turn	Medium	11.67190, 78.23144	30 KM/Hr
23	Turn	Medium	11.67429, 78.23896	30 KM/Hr
24	Turn	High	11.74622, 78.28147	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
9	hospital	Vinayaka Mission's Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
13	hospital	Thayammal Hospitals	11.6237472, 78.1177631	30 km/h	Medium
14	hospital	Vanitha Hospital	11.6239931, 78.1184622	30 km/h	Medium
16	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
17	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
18	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
19	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
20	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
21	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
22	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
23	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
24	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
25	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
26	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
31	hospital	M. Thathanur Primary Health Center	11.6824017, 78.2445813	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
10	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
11	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
12	school	Sri Vidhya Bharati Matric HR Sec School	11.6186069, 78.1103239	30 km/h	Medium
15	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
27	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
28	school	Kailash Manasarover School	11.6528379, 78.2021579	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
29	school	Holy Cross Matriculation Higher Secondary School	11.6635657, 78.203532	30 km/h	Medium
30	school	MettupattiThathanur Primary School	11.6827707, 78.2442509	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63266, 78.12699



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45358, 77.85781



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45353, 77.85697



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45853, 77.85139



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46307, 77.84941



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46316, 77.84921



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45631, 77.81668



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60942, 78.10056



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60965, 78.10070



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.62145, 78.10740



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.62130, 78.10776



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.61217, 78.10439



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63297, 78.12749



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62826, 78.13337



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.65632, 78.20220



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66380, 78.20648



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67111, 78.23134



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67190, 78.23144



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67429, 78.23896



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.74622, 78.28147
